

Optimal Climate Policy with Incomplete Markets

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University of Toronto, April 10, 2025

Motivation

- At some point, we will need to do something about climate change
- **Carbon taxes** are the obvious answer, but they can:
 - disproportionately burden poorer households
 - exacerbate existing inequalities
- Two questions:
 - Should governments temper climate ambitions in response to inequality?
 - Or should the broader tax system be adjusted to align climate action with distributional goals?

What we do

- Develop a fiscal climate-economy model à la [Barrage \(2020\)](#), with inequality and idiosyncratic labor-income risk à la [Aiyagari \(1994\)](#)

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- **Theoretically:**
 - Characterize the optimal carbon tax in an incomplete-market setting
 - Compare with the first-best Pigouvian benchmark
- **Quantitatively:**
 - Calibrate the model to match US macro data, inequality, and income risk
 - Solve the Ramsey problem to study:
 1. Optimal climate policy
 2. Its effects on the economy and welfare

What we find

- **Theoretically:**

- *First-best:* Pigouvian rule — carbon tax equals social cost of carbon (SCC)
- *Second-best:* SCC adjusted for marginal cost of public funds (MCF)
- *Third-best:* further deviations under instrument constraints

What we find

- **Theoretically:**

- *First-best:* Pigouvian rule — carbon tax equals social cost of carbon (SCC)
- *Second-best:* SCC adjusted for marginal cost of public funds (MCF)
- *Third-best:* further deviations under instrument constraints

- **Quantitatively:**

- Optimal carbon tax is essentially equal to the SCC
- Holds regardless of flexibility in other instruments
- True even when other tools have large economic effects
- Setting the carbon tax too low is far more costly than setting it too high

Contribution (1/2)

We contribute to three strands of literature:

- Optimal climate policy with distortionary taxation (e.g., [Bovenberg and de Mooij, 1994](#); [Jacobs and de Mooij, 2015](#); [Barrage, 2020](#); [Douenne et al, 2023](#))
 - **Our paper:** introduce incomplete markets
- Distributional effects of climate policy (e.g., [Känzig, 2023](#); [Fried et al, 2018 and 2023](#); [Benmir and Roman, 2022](#))
 - **Our paper:** study optimal policy, analyze the transition, and account for welfare benefits of mitigation
- Optimal fiscal policy with incomplete markets (e.g., [Conesa et al, 2009](#); [Dyrda and Pedroni, 2023](#))
 - **Our paper:** introduce climate change and study climate policy

Contribution (2/2)

Other papers on optimal carbon taxes in an Aiyagari-type models:

- [Bourany \(2024\)](#): cross-country heterogeneity; no other fiscal instruments
- [Kubler \(2024\)](#): constrained-optimal policy à la [Dávila et al. \(2012\)](#); theory-focused, abstracts from redistribution
- [Belfiori, Carroll, Hur \(2024\)](#): constrained-optimal policy; combines theory and quantitative analysis
- [Malafray and Brinca \(2022\)](#): early attempt at full welfare maximization; stylized (2 periods, no other instruments)
- [Wöhrmüller \(2024\)](#): richer model and careful calibration, but fixes other instruments and focuses on steady-state

➤ **Our paper:** solve for dynamic optimal fiscal policy to maximize social welfare, with a rich calibration and more flexibility over the instruments

Model

Model: Households

- Continuum of households, with preferences over consumption, labor, and temperature:
 $\mathbb{E}_0 \left[\sum_t \beta^t u(c_t, h_t, Z_t) \right]$
- Individuals characterized by assets $a \in A$ and stochastic productivity $e \in E$ that follows a Markov process with matrix Γ
- Given a sequence of prices and taxes the household solves

$$v_t(a, e) = \max_{c_t, h_t, a_{t+1}} u(c_t(a, e), h_t(a, e), Z_t) + \beta \sum_{e_{t+1} \in E} v_{t+1}(a_{t+1}(a, e), e_{t+1}) \Gamma_{e, e_{t+1}}$$

subject to

$$c_t(a, e) + a_{t+1}(a, e) = (1 - \tau_{H,t}) w_t e h_t(a, e) + (1 + (1 - \tau_{K,t}) r_t) a_t + T_t$$

$$a_{t+1}(a, e) \geq \underline{a}$$

Model: Firms

- Final good sector

$$Y_t = (1 - D(Z_t))A_{1,t}F(K_{1,t}, H_{1,t}, E_t)$$

- Energy sector

$$E_t = A_{2,t}G(K_{2,t}, H_{2,t})$$

- Energy production generates emissions $E_t^M = (1 - \mu_t)E_t$

➤ μ_t is the fraction of pollution abated at cost $\Theta_t(\mu_t, E_t)$

- With $\tau_{E,t}$ denoting carbon taxes, profits are

$$\mathcal{P}_t = p_{E,t}E_t - w_tH_{2,t} - (r_t + \delta)K_{2,t} - \tau_{E,t}E_t^M - \Theta_t(\mu_t, E_t)$$

Model: Government and Climate

- The government's budget constraint is

$$G_t + T_t + r_t B_t = \tau_{H,t} w_t H_t + \tau_{K,t} r_t (K_t + B_t) + \tau_{E,t} E_t^M + (B_{t+1} - B_t)$$

- Temperature Z_t determined by initial stock of carbon, $\mathcal{E}_0$, and history of endogenous and exogenous emissions:

$$Z_t = J(\mathcal{E}_0, E_0^M, \dots, E_t^M, E_0^{\text{ex}}, \dots, E_t^{\text{ex}})$$

Competitive equilibrium and Ramsey problem

- The **competitive equilibrium** is defined as usual (i.e., households and firms maximize given prices and policies, laws of motion are consistent, and markets clear) [► Details](#)

- **Ramsey problem:** Given equilibrium constraints, choose time path of policies $\pi \equiv \{\tau_{H,t}, \tau_{K,t}, \tau_{E,t}, T_t\}_{t=0}^{\infty}$ to maximize the (utilitarian) social welfare function,

$$\mathcal{W}(\pi) = \int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t(a_0, e_0 | \pi), h_t(a_0, e_0 | \pi), Z_t(\pi)) \right] d\lambda_0$$

- We impose additional constraints on $\{\tau_{H,t}, \tau_{K,t}, T_t\}_{t=0}^{\infty}$ and study effects on optimal $\{\tau_{E,t}\}_{t=0}^{\infty}$

Some theory

Optimal carbon tax formula (1/3)

- In the **first-best** (complete markets *and* individualized transfers), the optimal carbon tax follows a simple Pigouvian rule,

$$\tau_{E,t}^{FB} = \sum_{j=0}^{\infty} \beta^j \left(\frac{\mathcal{W}_{c,t+j}}{\mathcal{W}_{c,t}} D'_{t+j} A_{1,t+j} F_{t+j} - \frac{\mathcal{W}_{Z,t+j}}{\mathcal{W}_{c,t}} \right) J_{E_t^M, t+j}$$

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- When markets are incomplete and individualized transfers are not available, the **second-best** tax is

$$\tau_{E,t}^{SB} = \sum_{j=0}^{\infty} \beta^j \left(\frac{\nu_{t+j}}{\nu_t} D'_{t+j} A_{1,t+j} F_{t+j} - \frac{\mathcal{W}_{Z,t+j}}{\nu_t} \right) J_{E_t^M, t+j}$$

with ν_t the Lagrange multiplier on the resource constraint

Optimal carbon tax formula (2/3)

- The marginal cost of public funds captures how costly it is for the planner to raise one extra unit of revenue:

$$\text{MCF}_t \equiv \frac{\nu_t}{\mathcal{W}_{c,t}}$$

- Using this, the optimal tax $\tau_{E,t}^{SB}$ can be expressed as:

$$\tau_{E,t}^{SB} = \sum_{j=0}^{\infty} \beta^j \left(\frac{\text{MCF}_{t+j}}{\text{MCF}_t} \frac{\mathcal{W}_{c,t+j}}{\mathcal{W}_{c,t}} D'_{t+j} A_{1,t+j} F_{t+j} - \frac{1}{\text{MCF}_t} \frac{\mathcal{W}_{Z,t+j}}{\mathcal{W}_{c,t}} \right) J_{E_t^M, t+j}$$

- Deviations from Pigouvian taxation arise only when raising revenue is distortionary, $\text{MCF} \neq 1$

Optimal carbon tax formula (3/3)

- Assuming $\{\tau_{H,t}, \tau_{K,t}, T_t\}_{t=0}^{\infty}$ are optimized, we can show:

$$\nu_t = \mathcal{W}_{c,t} + \sum_{e_i^t} \Pr(e_i^t) (SD_{it} + LD_{it})$$

where

$$SD_{it} \equiv u_{cc,it}(\lambda_{it}^{EE} - R_t \lambda_{it-1}^{EE}) \quad (\text{Savings Distortions})$$

$$LD_{it} \equiv (u_{cc,it} w_t (1 - \tau_{Ht}) e_{it} + u_{ch,it}) \lambda_{it}^{LS} \quad (\text{Labor Distortions})$$

- $SD_{it} = 0$ if household is borrowing constrained
- $LD_{it} = 0$ if there are no income effects on labor supply
- Constraints on $\{\tau_{H,t}, \tau_{K,t}, T_t\}_{t=0}^{\infty}$ imply additional terms
- To say more about $\tau_{E,t}^{SB}$, we use computational methods

Computational method

- We want to find the time paths $\{\tau_{K,t}, \tau_{H,t}, \tau_{E,t}, T_t\}_{t=0}^{\infty}$ that maximize welfare
- If optimal paths are smooth over time, we can approximate them with polynomials as in [Dyrda and Pedroni \(2023\)](#) [► Details](#)
- Polynomial parameters $\rightarrow$ path of fiscal instruments $\rightarrow$ transition to new balanced-growth path $\rightarrow$ welfare
- Optimize welfare by choosing polynomial parameters
- Bypasses the need to rewrite the Ramsey problem recursively

Quantitative analysis

Calibration strategy

- Climate model of the world
- Economic model of the US: scaled up to match global GDP and emissions
 - Inequality and fiscal policy of the US
 - GDP of the world, but GDP per capita of the US
- Optimal policy of the US if they assume rest of the world follows same climate policy

Climate model

- We use the model of [Dietz and Venmans \(2019\)](#) to capture two key climate facts: (i) temperature responds quickly and permanently to emissions, and (ii) temperature rises linearly with cumulative emissions
- Global mean temperature Z_t evolves as:

$$Z_{t+1} = Z_t + \epsilon(\zeta \mathcal{E}_t - Z_t)$$

where $\zeta = 0.00045^\circ\text{C}/\text{GtCO}_2$, and $\epsilon = 0.5$ governs adjustment speed

- Cumulative emissions $\mathcal{E}_t$ increase with industrial and land-use emissions:

$$\mathcal{E}_{t+1} = \mathcal{E}_t + E_t^M + E_t^{\text{ex}}$$

- Initial values from [IPCC \(2021\)](#) and [Global Carbon Project \(2022\)](#):

$$Z_{2020} = 1.1, \quad \mathcal{E}_{2020} = 2390, \quad E_{2020}^M = 36, \quad E_{2020}^{\text{ex}} = 4$$

- Exogenous land emissions follow [DICE \(2023\)](#): fall 10% every 5 years

Climate damages

- We model production damages following [Dietz and Venmans \(2019\)](#):

$$D(Z_t) = 1 - \exp\left(-\frac{\alpha_1}{2} Z_t^2\right)$$

- This form yields a damage curve similar to [DICE 2023](#)
- $\alpha_1 = 0.01$ implies damages of 2% of GDP at 2°C, and 7.7% at 4°C

Households

- Preferences over consumption, labor, and climate:

$$u(c, h, Z) = \frac{(c^\gamma(1 - \varsigma h)^{1-\gamma})^{1-\sigma} + (1 + \alpha_z Z^2)^{\sigma-1}}{1 - \sigma}$$

- $\{\beta, \gamma, \sigma, \varsigma\}$ calibrated to match:
 - Capital-output ratio of 2.6 ([NIPA, 2009–2019](#))
 - Average hours worked = 0.24 ([CPS](#))
 - Intertemporal elasticity of substitution = 1/1.5
 - Average Frisch elasticity = 1.0
- α_z chosen so that 26% of damages directly affect utility ([Barrage, 2019](#))

Labor Productivity Process

- Labor productivity has two components:
 - Persistent component e_P : Markov process with 4 states
 - Transitory component e_T : i.i.d. with 6 values
- Combined as: $e = e_P + e_T e_P^\eta$
 - $\eta = 0$: additive shocks; $\eta = 1$: multiplicative
- 26 free parameters after normalizing average productivity to 1
- Parameters calibrated to match inequality, risk, and population structure

► Population Partition

► Parameters

Calibration: Inequality

Cross-sectional distributions

	Bottom (%) 0–5	1st	2nd	Quintiles 3rd	4th	5th	Top (%) 95–100	Gini
Wealth								
Data	−0.5	−0.5	0.8	3.4	8.9	87.4	65.0	0.85
Model	−0.2	0.1	1.7	3.6	6.7	88.1	70.0	0.85
Earnings								
Data	−0.1	−0.1	3.5	10.8	20.6	65.2	35.3	0.65
Model	0.0	0.1	3.6	12.0	17.7	66.6	37.5	0.65
Hours								
Data	0.0	2.7	13.8	19.2	27.9	36.4	11.1	0.34
Model	0.0	0.4	11.4	26.1	28.3	33.9	8.9	0.35

Calibration: Labor-Income Risk

Statistical properties of labor income		
	Target	Model
Variance of 1-year growth rate	2.33	2.32
Kelly skewness of 1-year growth rate	-0.12	-0.13
Moors kurtosis of 1-year growth rate	2.65	2.65

- Data for households from [Pruitt and Turner \(2020\)](#)

Production

- Cobb-Douglas technology for both sectors

$$F(K_{1,t}, H_{1,t}, E_t) = K_{1,t}^\alpha H_{1,t}^{1-\alpha-\nu} E_t^\nu,$$

with $\alpha = 0.3$, and $\nu = 0.04$ ([Goloso et al, 2014](#)), and

$$G(K_{2,t}, H_{2,t}) = K_{2,t}^{\alpha_E} H_{2,t}^{1-\alpha_E},$$

with $\alpha_E = 0.6$ ([Barrage, 2019](#))

- Abatement cost function adapted from [DICE 2023](#):

$$\Theta(\mu_t, E_t) = P_t^{\text{back}} \frac{\mu_t^{c_2}}{c_2} E_t,$$

where P_t^{back} is the backstop price

Government

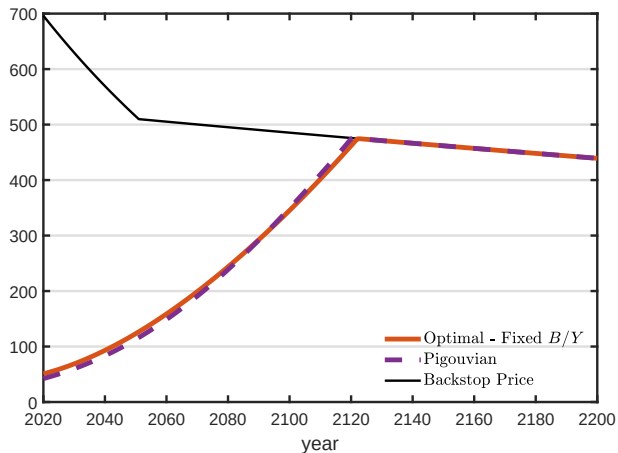
- We extend procedure of [Trabandt and Uhlig \(2012\)](#) up to 2019:
 - labor income tax rate: $\tau_{H,0} = 27.7\%$
 - capital income tax rate: $\tau_{K,0} = 33.6\%$
- Debt (2019): difference between total liabilities and financial assets from the US government's balance sheet: $B_0/Y_0 = 104.5\%$
- Transfers capture all personal transfers from NIPA: $T_0/Y_0 = 14.7\%$

Policy experiments

- We study a government that chooses a time-varying carbon tax under three fiscal regimes:
 1. Fixed debt-to-GDP, fixed other taxes
 - Transfers adjust to balance the budget
 2. Flexible debt-to-GDP, fixed other taxes
 - Transfers follow a flexible path: optimal to front-load transfers
 3. Flexible debt-to-GDP, optimal constant labor and capital taxes
 - Optimal to increase both taxes $\tau_H = 27.7\% \rightarrow 41.5\%$, and $\tau_K = 33.6\% \rightarrow 44.7\%$

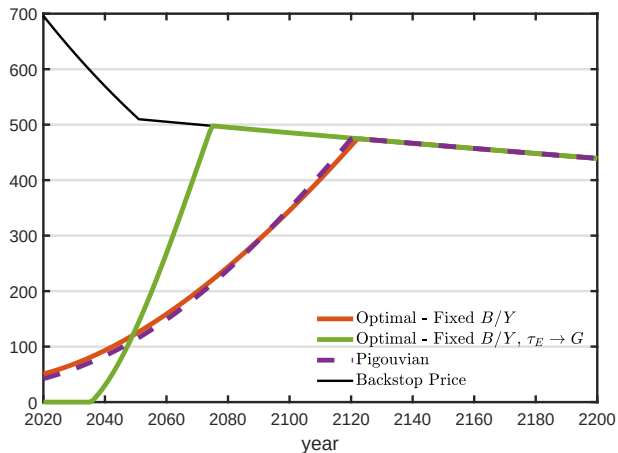
Results

Results: Optimal carbon tax (1/3)



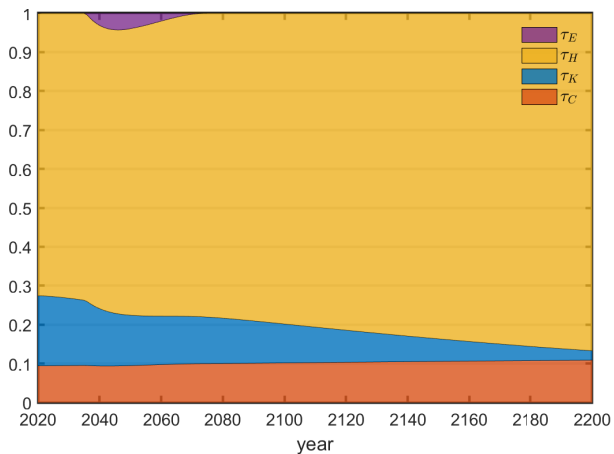
Optimal Carbon Taxes (in \$/tCO₂)

Results: Optimal carbon tax (intermezzo)



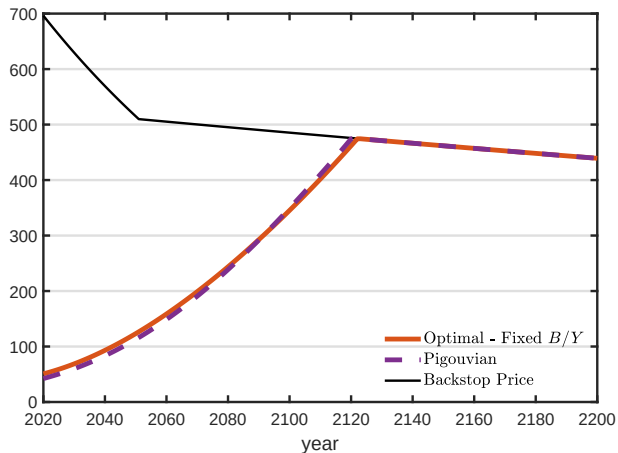
Optimal Carbon Taxes (in \$/tCO₂)

Results: Optimal carbon tax (intermezzo)



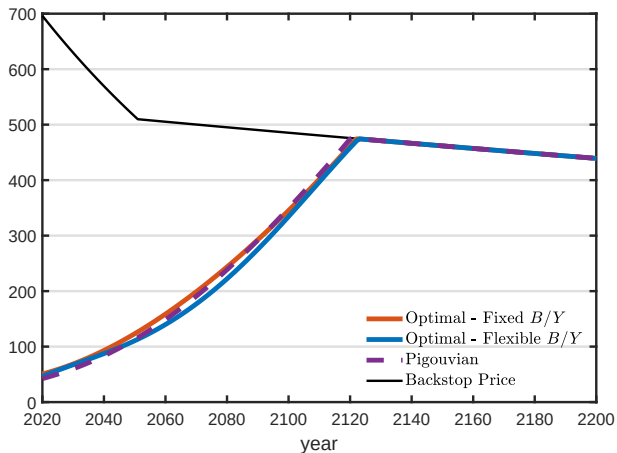
Revenue Shares

Results: Optimal carbon tax (1/3)



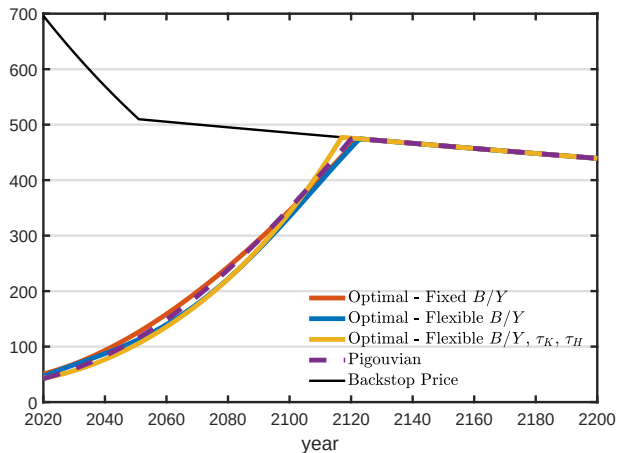
Optimal Carbon Taxes (in \$/tCO₂)

Results: Optimal carbon tax (2/3)



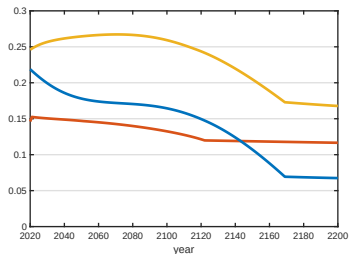
Optimal Carbon Taxes (in \$/tCO₂)

Results: Optimal carbon tax (3/3)

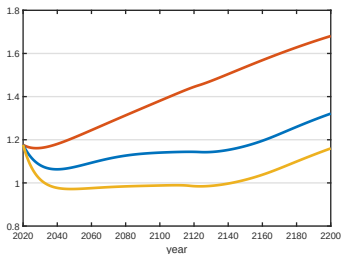


Optimal Carbon Taxes (in \$/tCO₂)

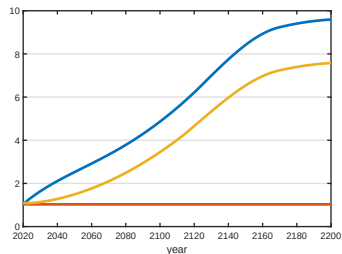
Results: Aggregates



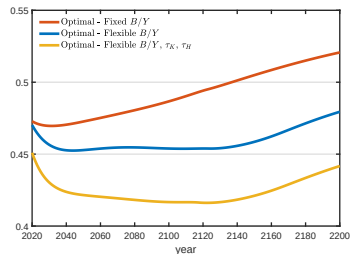
Transfers to Output



Capital



Debt to Output



Output

Results: Discussion

- Consider the second-best formula in period 0:

$$\tau_{E,0}^{SB} = \sum_{t=0}^{\infty} \beta^t \left(\frac{\text{MCF}_t}{\text{MCF}_0} \frac{\mathcal{W}_{c,t}}{\mathcal{W}_{c,0}} D'_t A_{1,t} F_t - \frac{1}{\text{MCF}_0} \frac{\mathcal{W}_{Z,t}}{\mathcal{W}_{c,0}} \right) J_{E_0^M, t}$$

- Front-loading transfers reduces $\mathcal{W}_{c,0}$, why does it not raise short-term carbon taxes?
1. Higher debt crowds out capital, reducing production damages:

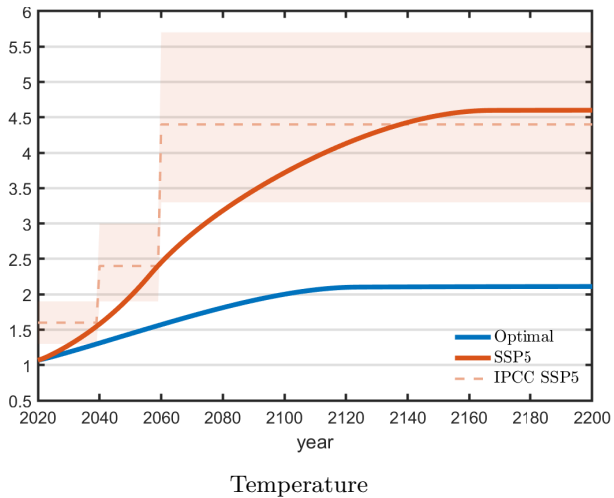
$$D'_t A_{1,t} F_t$$

2. Production damages represent 3/4 of overall damages and depend on growth trajectory, not the level of $\text{MCF}_t \mathcal{W}_{c,t}$

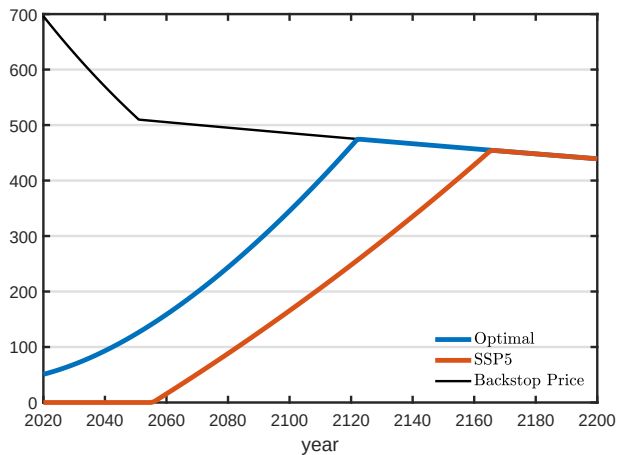
Welfare gains

- We want to compute the welfare gains associated with implementing the optimal policy
- A commonly used benchmark is the [IPCC \(2021\)](#)'s Shared Socioeconomic Pathway 5 (SSP5) scenario
- It is supposed to capture a somewhat pessimistic business-as-usual scenario

Temperature in BAU Scenario

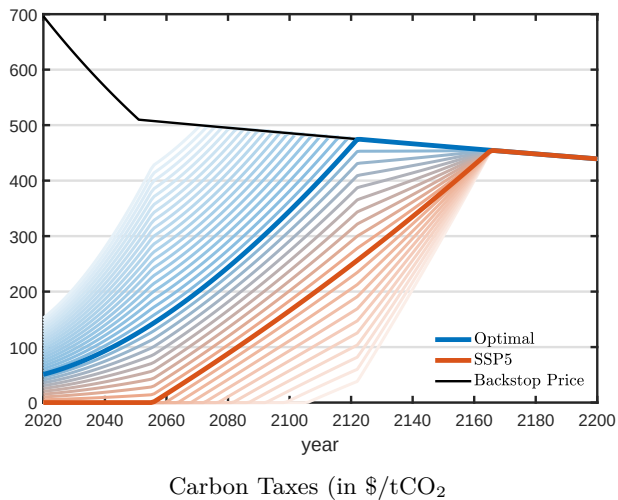


Corresponding Carbon Taxes

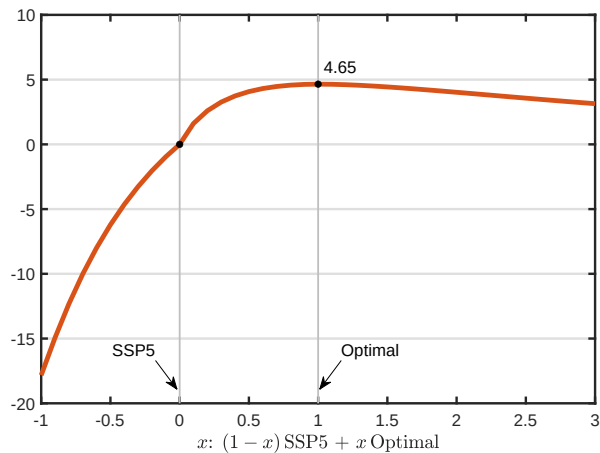


Carbon Taxes (in \$/tCO₂)

Varying Taxes From One to the Other



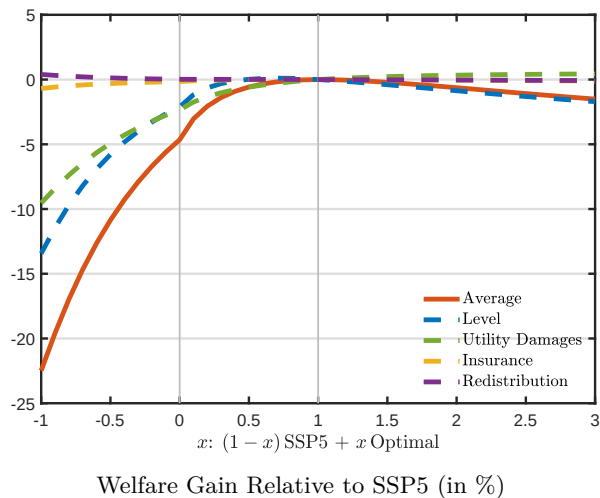
Welfare Gain



Welfare Gain Relative to SSP5 (in %)

- Doing too much not so costly, doing too little could very well be

Welfare Decomposition



- Insurance and redistribution are not significantly affected

Distribution of Gains

Asset Quintile	Q5	4.70	5.07	4.81	5.25	5.83
	Q4	4.59	4.60	4.58	4.58	4.59
	Q3	4.59	4.59	4.57	4.57	4.58
	Q2	4.58	4.58	4.57	4.57	4.57
	Q1	4.58	4.57	4.56	4.56	4.58
		Q1	Q2	Q3	Q4	Q5
		Labor Income Quintile				

Welfare Gain Relative to SSP5 (in %)

- Gains shared broadly, but richest households benefited most

Conclusion

- The optimal carbon tax is quantitatively very close to the social cost of carbon (SCC), even in the presence of fiscal distortions and policy constraints
- This result is robust across fiscal regimes and holds even when other instruments have large economic effects
- Setting the carbon tax too low imposes significantly higher welfare costs than setting it too high
- In the paper, we extend the model to include energy goods that are necessary, results are very similar though welfare gains are even larger

Appendix

Competitive Equilibrium: Definition

Given K_0, B_0, Z_0 , an initial distribution λ_0 , and a policy $\pi \equiv \{\tau_{H,t}, \tau_{K,t}, \tau_{E,t}, T_t\}_{t=0}^{\infty}$, a **competitive equilibrium** is a sequence of value functions $\{v_t\}_{t=0}^{\infty}$, an allocation $X \equiv \{c_t, h_t, a_{t+1}, Z_t, E_t, \mu_t, K_{1,t}, K_{2,t}, K_{t+1}, H_{1,t}, H_{2,t}, H_t, B_{t+1}\}_{t=0}^{\infty}$, a price system $P \equiv \{R_t, w_t, r_t, p_t^e\}_{t=0}^{\infty}$, and a sequence of distributions $\{\lambda_t\}_{t=0}^{\infty}$, such that for all t :

1. the allocations solve the consumers' and the firms' problems given prices and policies;
2. the sequence of probability measures $\{\lambda_t\}_{t=1}^{\infty}$ satisfies

$$\lambda_{t+1}(S) = \int_S Q_t((a, e), S) d\lambda_t, \quad \forall S \text{ in the Borel } \sigma\text{-algebra of } S, \quad (1)$$

where Q_t is the transition probability measure;

3. the government budget constraint is satisfied in every period, and debt is bounded;
4. temperature change satisfies the law of motion stated above in every period, and;
5. markets clear, i.e., the following equations are satisfied:

$$H_t = H_{1,t} + H_{2,t}, \quad (2)$$

$$K_t = K_{1,t} + K_{2,t}, \quad (3)$$

$$C_t + G_t + K_{t+1} + \Theta_t(\mu_t, E_t) = (1 - D(Z_t)) A_{1,t} F(K_{1,t}, H_{1,t}, E_t) + (1 - \delta) K_t, \quad (4)$$

$$E_t = A_{2,t} G(K_{2,t}, H_{2,t}), \quad (5)$$

$$H_t = \int_S e h_t(a, e) d\lambda_t, \quad (6)$$

$$K_t + B_t = \int_S a d\lambda_t. \quad (7)$$

The economy is on a **balanced growth path** if all the aggregate variables grow at a constant rate and the economy satisfies competitive equilibrium conditions. [◀ Back](#)

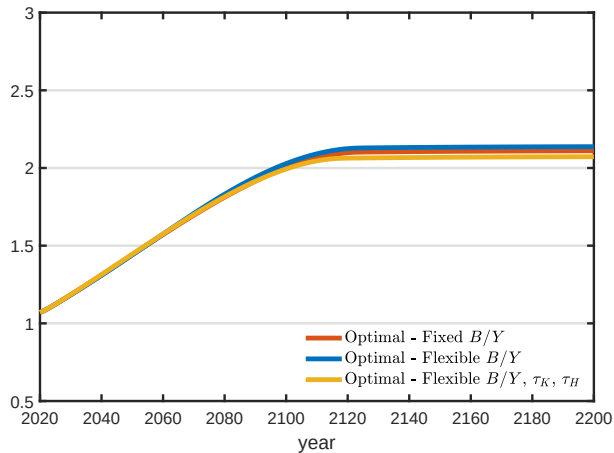
Computational Method: Details

- Solving this problem involves searching on the space of sequences $\{\tau_{K,t}, \tau_{H,t}, \tau_{E,t}, T_t\}_{t=0}^{\infty}$
- To reduce the dimensionality of the problem, we follow [Dyrda and Pedroni \(2023\)](#) and parameterize the time paths of fiscal instruments:

$$x_t = \left(\sum_{j=0}^{m_{x0}} \alpha_j^x P_j(t) \right) \exp(-\lambda^x t) + (1 - \exp(-\lambda^x t)) \left(\sum_{j=0}^{m_{xF}} \beta_j^x P_j(t) \right)$$

- x_t can be any of the fiscal instruments $\{\tau_{H,t}, \tau_{K,t}, \tau_{E,t}, T_t\}$
- $\{P_j(t)\}_{j=0}^{m_{x0}}$ and $\{P_j(t)\}_{j=0}^{m_{xF}}$ are Chebyshev polynomials
- $\{\alpha_j^x\}_{j=0}^{m_{x0}}$ and $\{\beta_j^x\}_{j=0}^{m_{xF}}$ are weights on the consecutive elements
- λ^x controls the convergence rate of the fiscal instruments

Temperature



Temperature Change

Calibration: Macroeconomic Variables

Macroeconomic aggregates		
	Target	Model
Intertemporal elasticity of substitution	0.66	0.66
Capital to output	2.57	2.54
Average Frisch elasticity (Ψ)	1.00	1.00
Average hours worked	0.24	0.25
Transfer to output (%)	14.7	14.7
Debt to output (%)	104.5	104.5
Fraction of hhs with negative net worth (%)	10.8	11.5
Correlation between earnings and wealth	0.51	0.43

Calibration: Population Partition

	Shares			
	Population	Earnings	Income	Wealth
Workers				
Data	67.2	82.7	69.1	44.9
Model	70.9	86.3	78.7	47.0
Business Owners				
Data	5.8	13.7	16.1	33.0
Model	6.6	13.7	14.8	31.2
Inactive Households				
Data	27.0	3.6	14.8	22.2
Model	22.5	0.0	6.5	21.8

Notes: Data comes from 2019 wave of the SCF

Labor Productivity Process: Parameters

Parameter				Value
Curvature η				1.12
Persistent shock		Transitory shock		
$\Gamma_P = \begin{bmatrix} 0.994 & 0.002 & 0.004 & 3\text{E-}5 \\ 0.019 & 0.979 & 0.001 & 9\text{E-}5 \\ 0.023 & 0.000 & 0.977 & 5\text{E-}5 \\ 0.000 & 0.000 & 0.012 & 0.987 \end{bmatrix}$		$e_P = \begin{bmatrix} 0.185 \\ 0.305 \\ 0.537 \\ 27.223 \end{bmatrix}$		
		$P_T = \begin{bmatrix} 0.357 \\ 0.002 \\ 0.467 \\ 0.004 \\ 0.025 \\ 0.176 \end{bmatrix}$		
		$e_T = \begin{bmatrix} 0.07 \\ 0.09 \\ 3.12 \\ 3.16 \\ 7.80 \\ 9.51 \end{bmatrix}$		

Welfare decomposition

- The utilitarian welfare function can increase for four reasons:
 1. Reduction in distortions, if the utility of the average agent, $\sum_{t=0}^{\infty} \beta^t u(C_t, N_t)$, increases: **the level effect** (Δ_L)
 2. Lower utility damages from climate, $\sum_{t=0}^{\infty} \beta^t v(Z_t)$: **the utility-damages effect** (Δ_{UD})
 3. Transfers from ex-post rich to ex-post poor, if the risk of each individual path $\{c_t, n_t\}_{t=1}^{\infty}$ is reduced: **the insurance effect** (Δ_I)
 4. Transfers from ex-ante rich to ex-ante poor, if the inequality between certainty equivalents for $\{c_t, n_t\}_{t=1}^{\infty}$ is reduced: **the redistribution effect** (Δ_R)

Proposition

Let Δ be the utilitarian (average) welfare gain. The following decomposition holds:

$$(1 + \Delta) = (1 + \Delta_L) (1 + \Delta_{UD}) (1 + \Delta_I) (1 + \Delta_R)$$